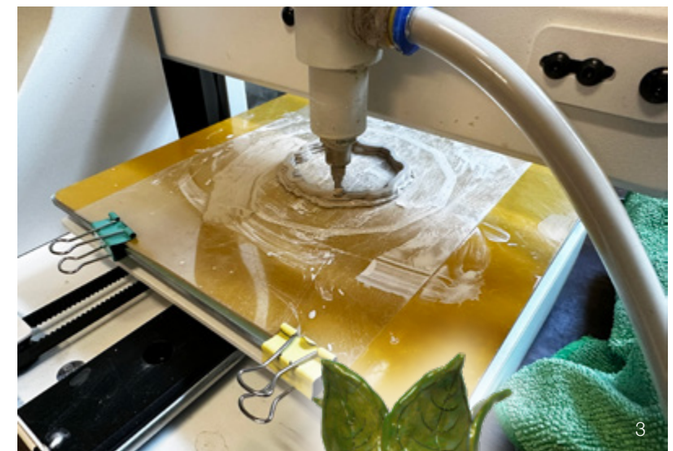
Kiichi Takeuchi



## Progress and Challenge: Scan and Print 3–D in a Classroom

In this article, I report on the progress, challenges, and evolution of the 3–D scanning and printing process at Long Island University.

When the department purchased the first clay 3–D printer in the beginning of this year, I was assigned to maintain, research, and utilize it because I had a Computer Science background. The product is an inexpensive model (Tronxy Moore 1). The mechanism is simple. I struggled for a few weeks because of the simplicity; many processes needed to be manually configured. After a while, I got better at handling technical problems and I was able to generate quality outcomes, so I started teaching and demonstrating to other people in the studio.

An instructor asked me if it is possible to clone his sculpture or not and I was motivated by his challenge. The reason is, we are good at building sculpture or pottery; however, we have few people who are familiar with building their own 3–D models on desktop computers in the craft center. I found it to be a great inspiration to use our own designs rather than printing from premade 3–D models. I spent time researching apps on my smartphone in order to find the best method to scan small objects. After a few trial and errors, I successfully scanned and printed from sculptures.

We have had *ad hoc* 3–D scanning and printing sessions with the group since spring 2022. There were many experiments for the original scanned sculptures vs. the 3–D printed ones by conducting different firing methods including the electric kiln, a microwave raku, and a wood-fired kiln (Anagama).

In the Fall 2022 semester, I introduced 3–D printing to a group of ten graduate students who were taught by a tenured faculty member, Professor Frank Olt. These students study wheel techniques, sculpture, and other technical topics. I introduced 3–D scanning and printing of their own artwork as a technical topic.

We had no student artwork to scan in the first week of this semester, so I went through a few scanning demonstrations using other available ceramic artworks. Because not all the students were knowledgeable about 3–D modeling software, we used simple scanner apps instead of complex desktop authoring tools.

Since the spring of 2022, I introduced the students to our 3–D printer (Tronxy Moore 1) and explained how it works. I clarified each part of the printer from the tube to the nozzle. I found more positive feedback on understanding the mechanism when I introduced the simplicity of the physical process rather than going over too many details of digital data format and so on. The nozzle moves horizontally and it draws outside the contour line with slips from the tube, and then it builds the structure vertically over time. Many students were already familiar with the process via a famous time-lapse video on YouTube where an entire house is printed; it was easy to explain that this is a miniature version of that.

Understanding the capability and limitations of 3–D printers was the key to the next step because students started thinking about selecting printable shapes from their artworks. Since it is drawing contour outlines with slip, the 3–D model has to be a self-contained structure; overhung walls or closed tops are difficult to print. I often use the following analogy from my colleague: "If you can make a sculpture by layering wet spaghetti, then you can print your 3–D model with this printer."

Our inexpensive 3–D clay printer has no heating system or powerful compressor to push out the slip from the nozzle. The alignment process is semi-manual; therefore, it requires some practice. Depending on the size of the object, the operator needs to stay with the printer for at least a half hour. I like to introduce the printer as if it is just like any other pottery tool, i.e., a pottery wheel or slab roller. We, as ceramic artists, must become accustomed to using the tools.

Next, I distributed step-by-step instructions and allowed students to work on their own. Some students brought artwork from the previous semester to scan, and we had a few good 3–D models to start with.

1 *Kiichi Takeuchi (right) demonstrates how to scan ceramics using a smartphone app in 3–D Scan Workshop*, 2022, Photo credit: Daniel Borghard
2 *Student Ashley Degraaff printing her scanned sculpture*, 2022, Photo credit: Kiichi Takeuchi
3 *Printer close up, Porcelain slip*, 2022, Photo credit: Kiichi Takeuchi
4 *Original Sculpture by Ashley Degraaff*, 2022, Stoneware, 23" × 10" × 10", Photo credit: Kiichi Takeuchi

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Students retained the 3–D models as well as their 3–D printed works for their own portfolios.

I demonstrated the 3–D printing process with a small conical-shaped sculpture, and the scanned original model was built with a coil method. For this first version, I printed it in actual size so that we could compare how the printer recreated the surface of the coils. Next, I drew everyone's attention to how the two sculptures differed or how they could differ. The original model, when 3–D printed, can be modified in different ways:

- Resizing
- Skewing
- Split/Slice
- Merging/Stacking
- Replicating

Stacking is an idea that came from one of the students. She figured out the printer can partially print sliced objects, and we discussed an idea to stack them like a totem pole. This idea also applies to the resizing method because the original sculpture that she scanned was more than three feet, and we reduced it to less than 10 inches. Her original sculpture contains multiple layers so it was natural for her to reconstruct it differently by stacking and reproducing each printed layer.

For objects that were larger than the printer size, I talked about the Casa Covida project by Ronald Rael, who was the keynote speaker at National Council on Education for the Ceramic Arts (NCECA) 2021. In that project, his team printed a house with three modules: bedroom, bathroom, and living room, and they are larger than the printer.[4] Stacking or assembling printed items to build larger sculptures could be another option for our students.

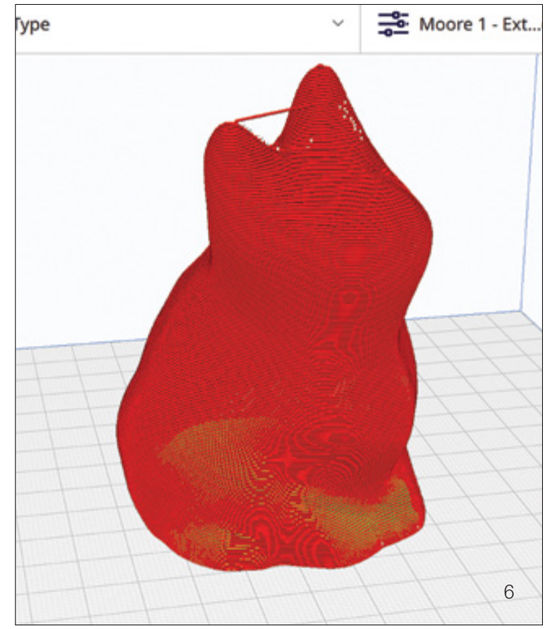
While working with the graduate student group, I initiated new activities to engage students in 3–D clay printing. I broadcast newsletters to the group about what is going on with other students' works and some related topics that this group might tackle by the end of the semester. I posted images of scanned items on Spatial.io, which is an interactive 3–D virtual reality platform, for other students to review.

We looked into projects and publicly available 3–D models at other universities. I wrote about the Shareware project and the Ceramic Commons project at Emmanuel College in Boston, Massachusetts.[4] The Shareware project is an ideal way to publish 3–D models of cups via the internet, and we printed a few cups from the website. We plan to publish our students' 3–D cup models online.

I scheduled a workshop about 3–D scanning for the general student body at the Craft Center. My aim is to attract other undergraduate students who are interested in building their digital portfolio using Spatial.io. Collecting 3–D models and curating them in Spatial.io is a great way to keep our institutional memory with those students' artwork. The virtual gallery could be useful for the next semester, and students can look them up as digital references.

I will be launching a competition for "The best scanned 3–D model award" with gift cards as the reward, and we are planning to continue this event every semester. Throughout those activities, my goal is to bring more students' attention to our 3–D scanning and printing learning process.



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Our initial study group contributed to our knowledge base by testing various scanning apps. We are not only improving the process by reviewing different apps, but we also tried to lower the cost for the entire life cycle of scanning and printing activity.

We started with a scanner app called Trnio. The cost of a Trnio perpetual license is $4.99USD. We tested other free apps, such as Polycam, WIDAR, Metascan, and Scaniverse. Since many people recommended Polycam, we started using it. It is free, and the app can export Graphics Language Transmission Format (GLTF).

Another app that we like is WIDAR. WIDAR which is also free, and most export options are available. In addition, WIDAR has an intuitive cropping tool compared to other apps, and it helps us to speed up the 3–D data-cleaning process so we can omit this post-processing step on our desktop computer. GLTF is a widely-available format in other platforms that we use in our group: Blender (3–D modeling software), Ultimaker-Cura (Slicing software), and Spatial.io.

As for the printing process, we keep improving and reducing the cost as well. I was inspired by K–12 educators who tried to use daily household products to improve the building process and lower the cost.[3] For example, disposable shrinkwrap is recommended to cover the print board; however, it tears apart when the nozzle scratches the surface, and it is not easy to transfer printed objects if it is still wet. Instead of wasting shrinkwrap, I cut some lids of thin plastic containers from a grocery store. Those thin clear plates improved transferring printed objects from print board to drying area, and I don't have to recalibrate the levels so often because we don't have to remove the entire board. To stabilize the printed wall, I use a regular hair dryer during the printing process, so that I can reduce the risk of a collapsed wall when the slip is too wet.

The next step for me is to reduce the cost of materials. I am trying to use recycled porcelain in the craft center because we are currently purchasing the clay pack from Tronxy. Although these costs are negligible in the short term, it is important to keep the running cost to a minimum level for our sustainable budget size so that we keep engaging students' motivations to learn new technology in ceramic education.

I use *A Guide to Clay 3D Printing* by Jonathan Keep for the ideal specific gravity of the slip, clay particle size, nozzle diameter vs. layer height ratio. The guide presents details of clay analysis and the results.[2]

**Kiichi Takeuchi** has a BS and MS in Computer Science from Long Island University. After developing apps and software for 23 years, Takeuchi became a BFA student in 2022. He is in charge of maintaining and leading the study group of a 3–D printer at the Craft Center.

**References:**

1. Carlisle, Mary. "Big Impact: Small Budget" YouTube, https://www.youtube.com/watch?v=Eflju61iSQ_NCECA. 2019.
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3. Naitoh, Megumi. "Share Ware Statement." Negumi Naitoh, http://www.meguminaitoh.com/shareware.
4. Rael, R., et al. *Casa Covida-Mud Frontiers III-Zoquetes Fronterizos III*. 40th Annual Conference of the Association for Computer Aided Design in Architecture: Distributed Proximities, ACADIA 2020.

5 **MASUMI YAMAGUCHI**, Left: *Original sculpture*, 2022, Stoneware, natural ash glaze, fired in an anagama kiln, 3" x 3" x 2" Right: *Printed version of the scanned sculpture*, 2022, Porcelain, Raku glaze, fired in Raku kiln, 5" x 5" x 5.5", Photo credit: Kiichi Takeuchi
6 *Scanned image imported in Ultimaker-Cura (Slicing software)*, Photo credit: Kiichi Takeuchi
7 **MASUMI YAMAGUCHI**, Scanned Original 3–D model, 2022, Photo credit: Kiichi Takeuchi